

## Case study: GradeMate

Anna, a senior data scientist at EduTech, is leading a project to develop an AI-based grading system for Danish high school essays. The goal is to reduce teacher workload by providing consistent, automated grading based on rubrics such as grammar, structure, and clarity. She and her team has named the model GradeMate. To build GradeMate, Anna's team used a pre-trained GPT-based language model, which was fine-tuned using a dataset of 35,000 anonymized essays previously graded by teachers.

The dataset contains exactly 5,000 essays for each of the 7 grades (-3, 00, 02, 4, 7, 10 and 12). These essays came from 25 high schools across the country, including 15 schools in high-income districts and 10 schools from lower-income areas. The dataset also represents a diverse range of students, including both native Danish speakers and non-native speakers.

Early pilot tests in a small number of schools showed a slight increase in students' homework completion rates after AI grading was introduced, although there is no clear evidence that this was due to the AI system itself. An optional feedback mechanism is included where teachers can alter the predicted grade and send this information back to EduTech for fine-tuning, but this feature has not been widely adopted in pilot programs.

In internal testing, Anna's team has found that, on the test set, GradeMate performs as indicated in the table below. The numbers are very similar for the train set.

| Metric               | Overall | Gender |        | District income |       | Native DK speaker? |      |
|----------------------|---------|--------|--------|-----------------|-------|--------------------|------|
|                      |         | Male   | Female | High            | Low   | Yes                | No   |
| Support              | 3,500   | 1,659  | 1,841  | 2,143           | 1,357 | 3,077              | 423  |
| Accuracy             | 0.96    | 0.94   | 0.98   | 0.96            | 0.95  | 0.97               | 0.92 |
| Recall (pass grades) | 0.87    | 0.86   | 0.88   | 0.87            | 0.87  | 0.91               | 0.61 |

Note that *support* is the number of data points in the test set belonging to the indicated category. *Accuracy* is the overall accuracy with 7 classes – one for each grade – while *recall (pass grades)* refer to the recall in the binary classification problem where pass grades (02, 4, 7, 10 and 12) are considered the positive class and fail grades (-3 and 00) are considered the negative class.

Anna is uncertain whether GradeMate is ready for deployment, and she needs to decide soon, as it's already November and to sell it to schools before the next school year, EduTech wants it done by January. You are to assist Anna in making her decision. Based on the information provided, **perform an ethical impact assessment and recommend how Anna should proceed**. Should GradeMate be deployed as is, or should Anna and her team take additional steps before launching – or should they abandon the project altogether?